NSE Stock Prediction Project

Methods tried, results, and what they mean

Report date: 12 October 2026 · Data: NSE daily prices, January 2015 to March 2025 (April 2025 onward held back, unused)

Summary

After testing about 200 rule variants and several types of model, no method predicts NSE large-cap stock returns well enough to claim an edge. Honest accuracy is 50 to 51.5% against a 50% chance level, and no excess return is statistically significant.

The project set out to predict Up, Sideways or Down for NSE stocks over the next 10 trading days and to beat a published base paper. That target has not been met.

The work did produce a result worth publishing. We measured four separate mistakes that each create a false "edge" on Indian stock data, and we rebuilt a recent published method and showed it has no predictive power under honest rules.

Question	Answer
Does any model beat simply holding the stocks?	No. The best signal, 12-month momentum, is +3.9% a year with a 90% interval of -2.2% to +9.8%.
Does machine learning beat simple rules?	No. Its apparent advantage came from training on future data and from a biased stock list.
Does the rebuilt comparison method work?	No. Next-day accuracy is 50.0 to 50.5%; it loses 2.6 to 5.1% a year after costs.
Was the data cleaning correct?	No. It wrongly removed 66 of 103 stocks. Now fixed: 95 are usable.
What can be published?	A replication and critique paper, with every number independently checked.

Timeline

Date (2026)	Work done
7 October	Rules fixed; first cleaning of the downloaded prices (103 symbols) and an audit of it
8 October	Labels, backtest engine, benchmarks, 35 indicator features; logistic regression and XGBoost; first trading backtest
9 October	Sector and long-term features; per-family models; RSI and EMA studies; portfolio tests on 30 stocks
11 to 12 October	Literature search; cleaning corrected; ranking models; published claims tested; comparison paper rebuilt; four independent checks

1. Project goal and original plan

The goal is a system that predicts whether an NSE stock will go Up, Sideways or Down over the next 10 trading days, for a final-year project and a journal paper. The stated target was to beat a base paper on accuracy or on profit after trading costs.

Planned design. A multi-agent system: a technical-indicator agent, a sector agent, a news agent using a language model, and a risk agent, combined by a stacking model. Only the technical and sector parts have been built. The news agent is blocked because historical news could not be downloaded.

Base paper. Kotekar, Mohan and Kolukuluri, "Can News Sentiment Improve Deep Learning Models for Nifty 50 Index Forecasting?", IEEE Access, 2026. It predicts the next-day direction of the Nifty 50 index using LSTM, GRU, SimpleRNN and TKAN models with news sentiment. Only its abstract and first page have been read; the full PDF is still needed.

Rules fixed at the start (7 October 2026)

Item	Setting
Stocks	NIFTY 100
Data	Daily prices from Yahoo Finance, January 2015 onward

Item	Setting
Development period	January 2015 to March 2025
Final test period	April 2025 onward, to be opened once at the end
Label	Up if price rises 2 x ATR first; Down if it falls 1 x ATR first; Sideways if neither within 10 days
Trading	Buy or stay in cash only; enter at the next day's open
Cost	0.35% per round trip (0.25% verified broker charges plus 0.10% assumed slippage)

Working rules. Report honestly, label every assumption, never state an unverified number, and pass a written check at every step before moving on. An older code folder (Multi_Agent) was set aside because its reported win rates were hardcoded text, not measured results.

2. System architecture of the current study

Figure 1 shows the pipeline used for the corrected study (sections 5 to 10). Each stage is a separate script, and a separate checker reviewed each stage before the next one started.

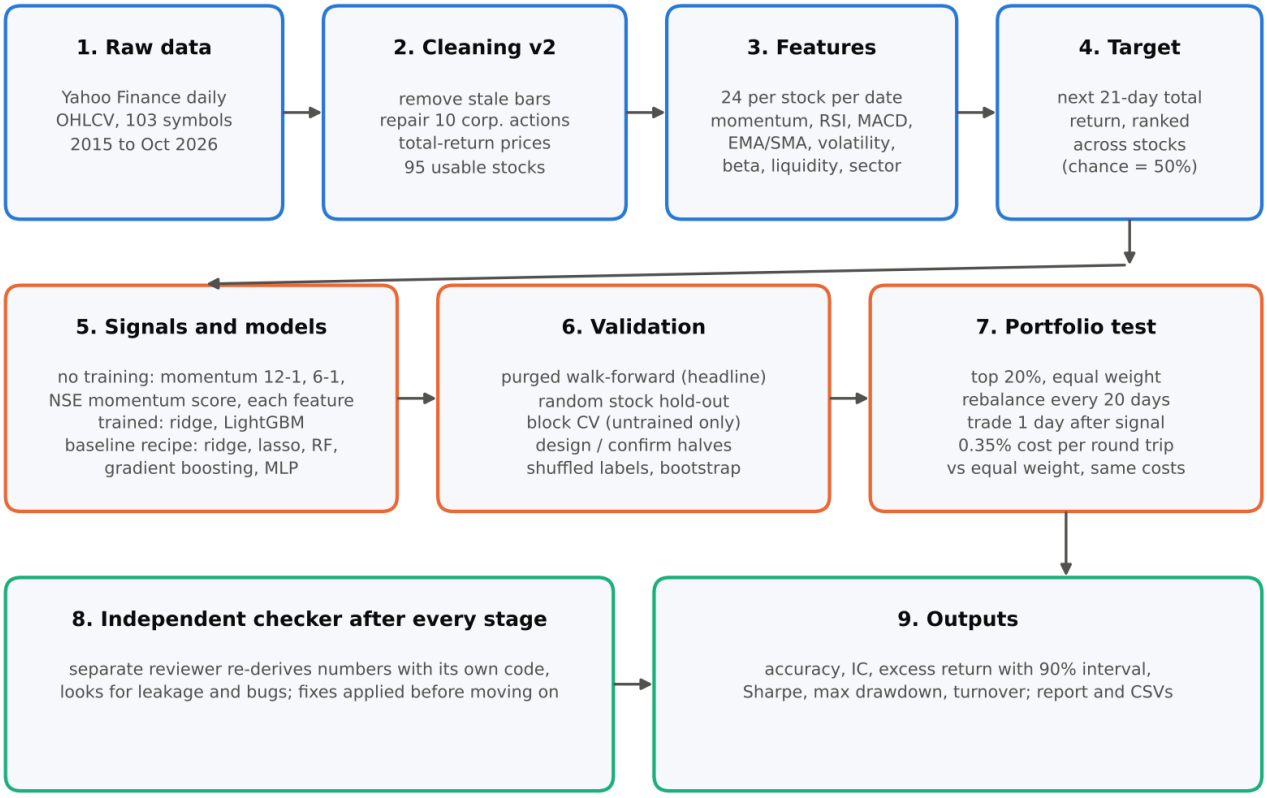


Figure 1. Pipeline of the corrected study. Blue: data preparation. Orange: modelling, validation and portfolio test. Green: checking and outputs.

3. What existed before this round (7 to 9 October 2026)

By 9 October the project had a working data pipeline, a backtest engine and many tested methods, and none of them beat simply holding the stocks after costs.

These earlier results used 8 pilot stocks (later 30) chosen by the old cleaning step, with price-only returns. They have not been re-run on the corrected data, so read them as history. Pilot stocks: TCS, RELIANCE, HDFCBANK, ITC, SIEMENS, TATASTEEL, BHARTIARTL, ULTRACEMCO. Out-of-sample window: January 2018 to March 2025.

Benchmarks to beat (after costs)

Benchmark	Return per year	Sharpe	Largest fall
Equal-weight hold of the 8 stocks	17.17%	0.98	-34.6%
NIFTY100 index hold	11.45%	0.72	-38.1%

Three-class prediction models (Up / Sideways / Down, 10 days ahead)

Method	Result	Pros	Cons
11 single-indicator rules (momentum, trend, volatility, volume)	None beat the benchmark. Best: EMA 20/50 trend, Sharpe 0.87	Simple, transparent	No standalone edge
Logistic regression, 35 indicator features	Macro-F1 0.335 vs 0.237 for always guessing the majority class; accuracy 48.8%	Fast floor model	Probabilities poorly calibrated
XGBoost, class-weighted	Macro-F1 0.371; accuracy 40.2%	Better balance across the 3 classes	Lower raw accuracy; still miscalibrated
XGBoost with isotonic calibration	Calibration error 0.081 to 0.042, but macro-F1 fell to 0.254	Trustworthy probabilities	Almost stops predicting Up or Sideways
Shuffled-label test	Noise floor 0.311, so the real edge is about 0.06 macro-F1	Confirms no leakage	Shows the edge is small
Trading the XGBoost signal (buy only when confident)	0.84% a year, Sharpe 0.68, largest fall -2.0%; in the market 2.8% of days	Very low risk	Loses badly to holding
45 features (added sector and long-term)	Macro-F1 0.370 (no change); Sharpe 0.18; 90% interval -0.45 to 0.88	Tests the "more features" idea	No improvement
One model per indicator family	Volatility-only best: Sharpe 0.53, 1.5% a year. Momentum, trend, volume, sector models negative	Shows which family carries signal	Still loses to holding

RSI and EMA rule studies

Method	Result	Pros	Cons
11 ways of using RSI	Trend-style RSI (stay long above 40) Sharpe 0.83; classic oversold bounce 0.14	RSI works far better as a trend filter	Still below holding (about 1.0)
RSI 60/40 rule alone	Sharpe 0.74; largest fall -20.4% vs -34.6%	Beat every ML model; smaller falls	Below equal-weight holding
Volatility model combined with RSI	Sharpe 0.40 to 0.52		Combining made it worse than RSI alone
RSI exit levels (19 variants)	Best exit is "almost never sell" (RSI below 20): Sharpe 1.009 on 18 trades vs 1.017 for holding	Smaller fall (-27.8%)	No better than holding
M and W chart patterns	W entry helped before costs, hurt after; M exit hurt in both cases		Costs remove the benefit
Random search over 60 RSI settings	Best in 2015-2020 (Sharpe 1.22) ranked 27th of 60 in 2021-2025	Honest overfitting test	"Best settings" are mostly luck
Grid of 108 RSI and EMA combinations	None beat each stock's own yearly return on average; best -0.69 points a year	Longer RSI (20) beats short (7)	52% win rate fell to 30% on unseen years
Portfolio changes on 30 stocks	More stocks raised Sharpe 1.02 to 1.22; volatility targeting, inverse-volatility weights and a market filter all hurt	Diversification helps	Holding the same 30 stocks scored 1.24

What this taught us. The rules that did best were the ones that traded least. Signals that time when to be in or out of a single stock lose to holding during a long rising market, because time spent in cash misses gains.

4. What we wanted next, and why the approach changed

The new aim on 11 October was modest: beat at least the weakest published stock-prediction paper on accuracy or profit, so the work could be published. Three changes followed.

1. **From timing one stock to ranking many.** Stronger published studies rank stocks against each other and stay fully invested, which removes the cash-drag problem. New task: on each date, rank stocks by expected return over the next 21 trading days.

2 **A second, weaker comparison paper.** Iskakov et al. (2026), "Machine Learning-Based Portfolio Optimization Using Cross-Sectional Signals: Evidence from the NSE Market", ETASR 16(4). 20 NSE large caps, eight indicator inputs, eleven models; reports 51 to 55% direction accuracy and Sharpe 0.95, with no trading costs and one train/test split.

3 **Re-examine the data cleaning**, because only 32 of 103 stocks were passing it.

Two further requests shaped the work: validation should not rely only on the last rows or last stocks, and claims found on the internet should be tested on our own data before being trusted.

New definition of accuracy. A prediction is correct if the stock is placed on the right side of that date's median return, so chance is exactly 50%. This differs from the earlier three-class accuracy and from the papers' next-day up/down accuracy, so numbers are not directly comparable across sections.

5. Data cleaning: what was wrong and what was fixed

The old cleaning step threw away 66 of 103 stocks for the wrong reason; the corrected cleaning keeps 95.

Check on the raw data	Finding	Action
Are share splits and bonus issues adjusted?	Yes. 79 split and bonus dates checked; none unadjusted. An earlier audit had claimed the opposite without reading the data	None needed
Are dividends handled?	Yahoo's dividend factor matched the reference table on all 1,404 matched dates, but the old clean files ignored dividends in returns	Built total-return prices (dividends are about 60% of total return for COALINDIA)
One-day moves of 15% or more	243 found. Almost all real: 25 on 23 March 2020 (COVID crash), 18 on 4 June 2024 (election result), PSU banks on 25 October 2017	Kept 234 as real history
The old filter	Excluded any stock with one such move. SBIN, ICICIBANK, INFY, LT, MARUTI were dropped for a single real crash day	Removed the filter
True corporate-action errors	10 dates, mostly demergers that Yahoo did not adjust	Repaired these 10 only
Stale bars	713 flat, zero-volume rows, mostly on exchange holidays	Removed

The 10 repaired dates. ADANIENT 3 June 2015, 5 April 2018 and 6 September 2018; CGPOWER 15 March 2016; ABB 20 December 2019; MOTHERSON 14 January 2022; ITC 6 January 2025; and TMPV, TRENT and VEDL in the held-back period. For ITC the true holder return is known; for the others that day's return is set to the index return, an estimate.

Why this matters. Keeping only stocks that never had a large move selects calm stocks using knowledge of their future. The errors themselves also create false profit: repairing two ADANIENT dates alone cut one model's apparent advantage from +7.1% to +4.8% a year.

Not fixable. Yahoo has no stock data for the real session of 18 March 2025. Some open, high and low prices contain bad ticks, so the new models use closing prices and volume only.

6. New methods tried on the corrected data

On 95 stocks, every ranking method scored between 50.0% and 51.5% accuracy, and none beat equal-weight holding by a statistically significant margin.

Setup. On every fifth trading day each stock gets a score. The portfolio holds the top 20% by score in equal amounts, rebalances every 20 trading days and trades one day after the signal. Cost is 0.35% per round trip. The benchmark is an equal-weight hold of the same stocks, paying the same costs. Models are trained only on data before each test year (yearly retraining, 2018 to March 2025). "Excess" is return per year above the benchmark.

Inputs. 24 features from closing price and volume: momentum over 3, 6 and 12 months; 1-week and 1-month reversal; volatility; RSI 14 and 20; EMA and SMA ratios; MACD; Bollinger position; distance from the 52-week high; beta; liquidity; sector-relative momentum. Each is ranked across stocks on each date.

Method	Accuracy	Excess / yr	90% interval	Sharpe	Pros	Cons
12-month momentum (no training)	51.5%	+3.9%	-2.2 to +9.8	1.23	Simplest; same sign in every test; low turnover	Not significant; bigger falls (-40%)
6-month momentum (no training)	51.1%	+4.7%	-0.9 to +10.5	1.30	Highest return of the simple signals	Not significant
Ridge regression, 22 features	50.0%	+1.1%	-4.0 to +5.9	1.22	Stable, interpretable	No edge
LightGBM, 22 features	50.8%	-1.2%	-6.4 to +3.7	1.05	Learns non-linear patterns	Worse than momentum; higher turnover
Ridge, all 24 features	50.7%	+3.4%	-2.7 to +9.3	1.28*		Gain comes from two liquidity features (section 9)
LightGBM, all 24 features	51.0%	+5.4%	-0.4 to +11.2	1.33*	Best-looking honest number	Same liquidity concern; not significant

Benchmark Sharpe 1.30 (rows marked * : 1.26, because those figures average four separate runs instead of one combined portfolio).

Same methods on a fairer stock list (rebuilt on each date as the 50 most-traded stocks at that time). The advantage disappears.

Method (top 50 by traded value)	Accuracy	Excess / yr	90% interval	Sharpe (benchmark 0.86)
12-month momentum	51.2%	-0.5%	-7.0 to +6.2	0.66
6-month momentum	50.7%	-2.7%	-9.4 to +3.8	0.58
Ridge, 22 features	50.5%	-2.2%	-7.5 to +2.9	0.66
LightGBM, 22 features	50.8%	-1.7%	-8.1 to +4.3	0.63

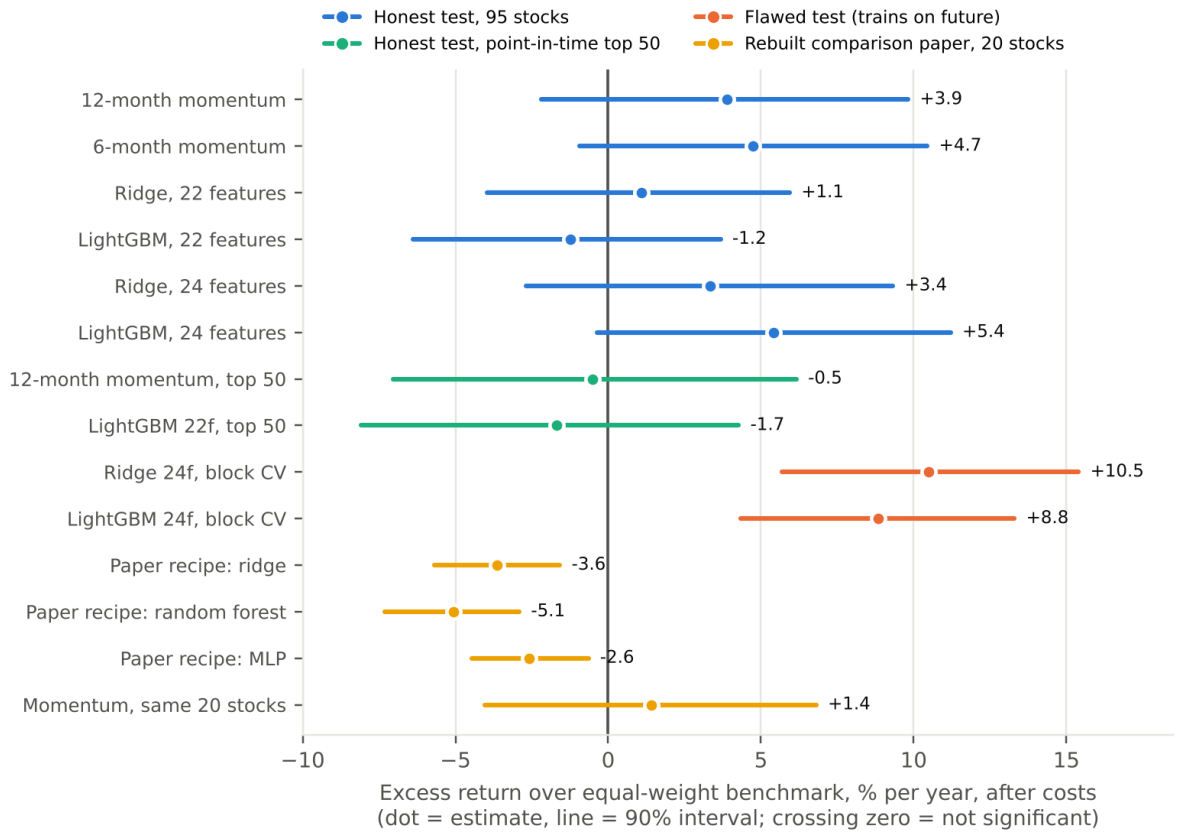


Figure 2. Excess return over the equal-weight benchmark, after costs, with 90% bootstrap intervals. Only the two orange rows exclude zero on the positive side, and both come from a flawed test that trains on future data (section 9). The yellow rows are the rebuilt comparison paper (section 8).

First test, for the record. The first version of the model test used 180 random combinations of held-out time blocks and held-out stocks. It gave accuracy of 51.2% for momentum, 50.4% for ridge and 50.2% for LightGBM, against 50.2% for a model trained on shuffled labels. It was later found unsafe for trained models (section 10), so the walk-forward tables above are the ones to rely on.

Models not run. LSTM and transformer models were not run in this round. PyTorch could not be installed in the analysis environment, and with signals this weak a larger model is unlikely to help.

7. Indicator accuracy over time, and published claims tested

No indicator stays above chance for long: every signal's accuracy drifts between about 46% and 55% from year to year, and the long-run averages sit within 1.6 points of 50%.

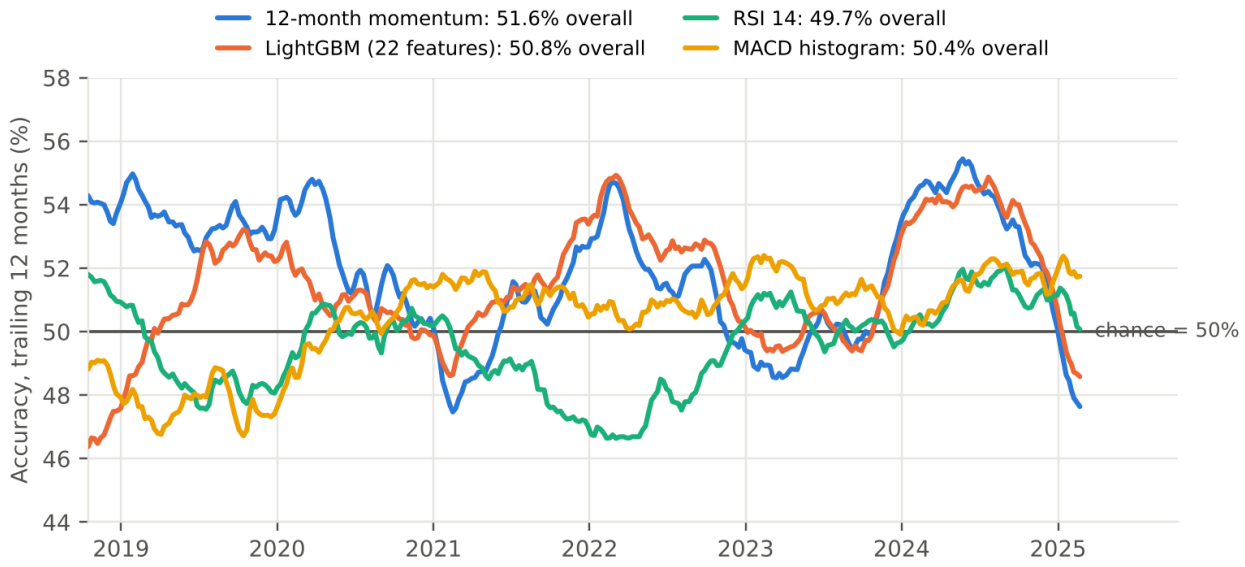


Figure 3. Accuracy of four indicators at ranking the 95 stocks (above or below the median 21-day return), measured over a trailing 12-month window, January 2018 to March 2025. LightGBM is trained on past data only. The horizontal line is chance. Overall accuracy for each is given in the legend.

The history was also cut into 10 time blocks, split at random into a "design" half and a "confirm" half. A claim counts as supported only if it holds in both. IC is the rank correlation between a signal and the next 21-day return.

Claim from the literature	Result on our data	Verdict
Past winners keep winning (momentum)	IC +0.030 and +0.035 in the two halves; accuracy 51.5% at 21 days, 52.4% at 63 days	Right direction, not significant
NSE's momentum-index formula is better than plain momentum	IC +0.034 and +0.017	No better
Low-volatility stocks outperform	IC +0.035 in one half, -0.079 in the other	Unstable; depends on market phase
RSI, MACD, Bollinger Bands and EMA crossovers identify winners	Accuracy 49.5 to 50.1%	No effect
Machine learning beats simple signals	Looked true (+10.5% a year) only with a flawed test (section 9)	Not supported
Trading only the most confident calls raises accuracy	Top and bottom 20% of calls: 51.3 to 51.8%	Very small gain
Choosing the best signal on past data works	The signal that looked best in the design half (beta) reversed in the confirm half	Not supported

Horizon. Momentum's accuracy rises with the holding period: 51.1% at 5 days, 51.5% at 21 days, 52.4% at 63 days. **Caution.** 29 signals were tested without a multiple-testing correction, so any single result near the significance threshold means little.

8. The comparison paper rebuilt and tested

Rebuilt under honest rules, the method of Iskakov et al. (2026) shows no ability to predict next-day direction and loses to equal-weight holding after costs.

Their recipe. 20 NSE large caps. Eight inputs (log return, RSI 14, MACD, ATR 14, Bollinger width, OBV, SMA 14, volume), scaled to 0-1. A 60-day window predicts the next day's return, one model per stock. Forecasts are averaged over the last 10 days, ranked across stocks and used as portfolio weights, rebalanced daily. No costs; one 80/20 split.

Our rebuild. Same 20 stocks and inputs, with ridge, lasso, random forest, gradient boosting and a small neural network (MLP). The scaler is fitted on training data only. Walk-forward over January 2018 to March 2025 (1,789 trading days), retrained each year.

Measure	Rebuilt method (range over 5 models)	Our 12-month momentum, same 20 stocks
Next-day direction accuracy	50.0 to 50.5% (always guessing "up" scores 51.3%)	Not applicable (monthly method)
Excess per year, their rules (no cost, no delay)	-0.5% to -3.6%	
Excess per year, 0.35% cost and one-day delay	-2.6% to -5.1%; every 90% interval below zero	+1.4% (interval -4.0% to +6.8%)
Sharpe after costs (equal-weight holding: 1.00)	0.67 to 0.82	0.96

Head to head, same days	Our lead per year	Probability we are ahead
Each method traded its own way, with costs	+4.1% to +6.5%	87 to 96%
Both with zero costs	+2.1% to +5.3%	73 to 92%
Their forecast traded monthly like ours, with costs	+1.1% to +9.2%	60 to 97%

About half of our lead comes from trading monthly instead of daily, which costs less. As a pure signal, ours is clearly ahead only of their random-forest and gradient-boosting versions.

- LSTM, their best model, was not run. This is a rebuild of the recipe, not a replication of their exact numbers.
- Their test period (mid-2024 to December 2025) lies in our held-back data, so the same dates could not be used.
- On their short 80/20 split (261 test days to March 2025) our momentum lost 9.9% to equal-weight and did worse than their models.
- Neither method beats equal-weight holding of the 20 stocks.

9. Four ways a false edge appears, each measured

Each of the first three mistakes, alone, was enough to make a model look profitable on our data, and the fourth made results look safer and larger than they were. This is the main contribution of the project so far.

Mistake	What it is	Measured effect on our data
Survivor-only stock list	Using today's NIFTY 100 members for the whole history. Companies that shrank or left are missing, so every stock is a known success	Equal-weight of today's members returned 24.6% a year (2018-2025); the NIFTY100 index 11.6%. On a list rebuilt each date (top 50 by traded value) the benchmark is 16.1%, and momentum's +3.9% becomes -0.5%
Training on the future	Testing on a middle period while training on later periods (block cross-validation)	Ridge and LightGBM showed +10.5% and +8.9% a year with intervals above zero. Trained on the past only: +3.4% and +5.4%, intervals include zero
Liquidity features on a survivor list	Features marking a stock as thinly traded; on a survivor list such stocks are small firms already known to have grown large	The whole ML advantage came from 2 of 24 features. Without them: -1.5% to +3.9% a year. Frequent "illiquid" picks include TRENT (38x its 2016 value) and SOLARINDS (19x)

Mistake	What it is	Measured effect on our data
Uncorrected data errors and loose measurement	Unadjusted demergers left in prices; drawdown read only every 20 days; costs ignored	Two ADANIENT dates added 2.3 points a year to one model. Largest fall understated by about 10 points (-27% vs the true -37%). Costs take 1.6 to 3.0 points a year from a daily-trading method

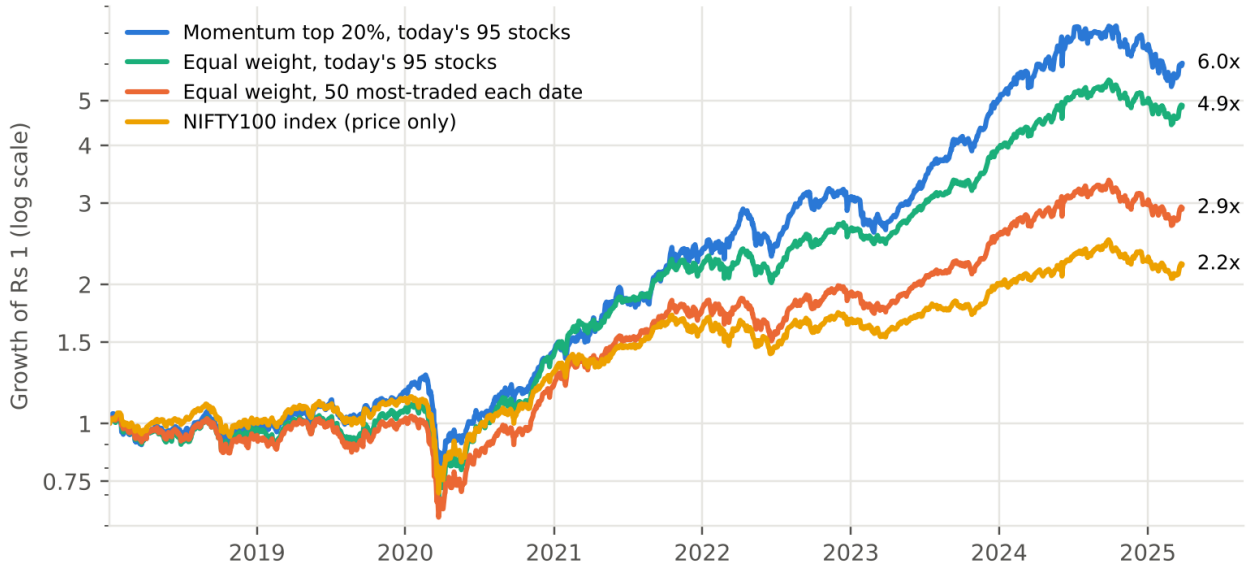


Figure 4. Growth of Rs 1, January 2018 to March 2025, after costs (index line: price only, no costs). The gap between the green line (today's 95 stocks) and the orange line (the 50 most-traded stocks on each date) is mostly survivorship bias, and it is larger than the gap that momentum adds on top.

A note on the third item. It cannot be proved that the liquidity effect is purely a survivor artefact, because our list contains no dropped companies. The evidence points that way: the signal works when a stock is compared with its own liquidity over the full sample (IC 0.088) and vanishes when compared with its own past only (IC 0.003).

Why this matters for published work. The comparison paper uses a fixed list of 20 current large caps, no costs and one split. Our measurements show how large the resulting overstatement can be on Indian data.

10. How the work was validated and checked

Every stage of the new work was reviewed by a separate automated checker that had not seen the work being produced, and its corrections were applied before moving on.

Method	What it does	Where used	Outcome
Purged walk-forward	Train on the past, test on the next year, with a gap so labels do not overlap	All headline numbers	The trusted method
Random stock hold-out	Test on stocks the model never trained on	First model test and claim tests	Kept
Combinatorial purged cross-validation	Test on every pair of time blocks: 45 combinations, 180 runs per model	First version of the model test	Dropped for trained models (lets a model train on later periods; added 5-7 points a year of false edge). Kept for untrained signals
Shuffled-label test	Scramble the answers; accuracy should fall to chance	Old and new models	Passed: 50.2%
Block bootstrap	Resample the return history 4,000-5,000 times to get an interval	All excess-return figures	Gives the 90% intervals quoted
Design and confirm halves	Choose on one random half of time, verify on the other	Claim tests	Picking the best signal does not carry over
Four start days	Average over the four possible rebalance start days	New portfolio results	Needed: result moved by 3 points by start day alone

Stage checked	Confirmed correct	Errors found and fixed
1. Cleaning	Splits, bonuses and dividends	3 missed demerger dates; 45 real bars deleted as stale; gaps in the index series
2. Model code	No look-ahead in features, targets, purging or costs; checker rebuilt momentum and benchmark with separate code and matched to four decimals	Drawdown understated; yearly returns overstated by about 0.6 points; result sensitive to start day
3. Claim tests	Backtest matched a hand-worked example exactly; all tables reproduced	Block cross-validation trains on the future; survivorship explanation stated too strongly
4. Paper rebuild	No leakage; scaler, windows and timing correct	Paired test misaligned by one day; a calendar gap briefly put the whole portfolio in one stock; our lead partly a cost effect

The request to avoid testing only on the last rows was met for stocks (random hold-out) and for untrained signals (random time blocks). For trained models it could not be met safely, for the reason in the first table.

How the analysis was produced. The code, runs and checks in this round were carried out with an AI assistant (Claude) under the students' direction. Every number in sections 5 to 9 comes from scripts saved in the project folder and can be re-run.

11. Limitations and what was not done

The conclusions hold for large NSE stocks, daily price and volume data, and 2015 to March 2025; they say nothing yet about mid caps, news or fundamentals.

- **Survivor-only list.** All 95 stocks are current NIFTY 100 members; no list of past members with dropped companies has been found. This inflates every return level in sections 3, 6 and 8.
- **Held-back data unused.** April 2025 onward has not been opened, so there is no final out-of-sample test. That period uses the same survivor list.
- **Earlier results not re-run.** Section 3 used the old cleaning and 8 or 30 stocks.
- **No LSTM or transformer in the new round.** The LSTM code for the paper rebuild is written but untested.
- **Original base paper not replicated.** The Kotekar et al. comparison needs the full paper and historical news data.
- **No news-agent or risk-agent results.** Two of the four planned agents are not built, and the stacking model that would combine them has not been attempted.
- **Estimated repairs.** Nine of the ten corporate-action repairs use the index return as an estimate of the true return.
- **One data source.** Prices come from Yahoo Finance only and were not cross-checked against NSE files.
- **Wider stock list not yet tested.** A NIFTY 500 download script is written but has not been run.
- **Statistical power.** About 7 years of test data on about 90 stocks can detect only a large edge; a small real edge could exist and remain unproven.

12. Proposed paper and next steps

The recommended paper is a replication and critique: it shows, with measurements on NSE data, how common shortcuts produce false trading results, and that a recently published method does not survive honest testing.

What the paper can claim

- The rebuilt comparison method has no next-day predictive power and loses to equal-weight holding after costs.
- A survivor-only stock list, training on future periods, and liquidity features each create a large false edge, with the sizes in section 9.
- Among 29 price and volume signals, only momentum has a consistent sign, and it is not statistically significant on large caps.

- Common technical indicators (RSI, MACD, Bollinger Bands, EMA) do not rank NSE large caps better than chance at a 21-day horizon.

What it cannot claim

- That any model here beats holding the stocks.
- That our method beats the comparison paper as a predictive signal.
- That the results hold on unseen data.

Next steps, in order

1. Run the LSTM version of the paper rebuild on a machine with PyTorch, so the comparison includes their best model.
2. Download the NIFTY 500 and repeat the tests on the top 100 and 200 stocks by traded value at each date. This removes most of the survivor problem and tests a wider set of stocks; the result could go either way.
3. Look for a historical index-membership list that includes dropped companies.
4. Obtain the full Kotekar et al. paper and decide whether the news-based comparison is still in scope.
5. Open the held-back period once, at the very end, for a single final test.

Decision needed from faculty. Whether a replication and critique paper is an acceptable outcome for the project, or whether a positive prediction result is required. The second is not supported by the evidence so far.

13. Glossary, files and sources

Term	Meaning
Accuracy (new task)	Share of stocks placed on the correct side of that date's median return; chance is 50%
Macro-F1	Average of the F1 score over the three classes, so a rare class counts as much as a common one
Excess per year	Strategy return minus benchmark return, per year, after costs
90% interval	Range that would contain the true value 9 times in 10; if it includes zero, the result is not statistically significant
Sharpe ratio	Return divided by its variability; higher is better
Largest fall (max drawdown)	Biggest peak-to-trough loss during the test
IC	Rank correlation between a signal and the following return
Momentum 12-1	Return over the past 12 months, skipping the latest month
Walk-forward	Train on the past, test on the next period, then roll forward
Purging	Dropping training rows whose outcome window overlaps the test period
Survivorship bias	Error from studying only companies that still exist or still qualify today
Total return	Price change plus dividends
ATR	Average true range, a measure of a stock's usual daily movement

Item	Location in the project folder
Technical report for the new work	reports/phase5/checkpoint_5_report.md
The four checker reports	reports/phase5/checker_report_s1.md to s4.md
Result tables	reports/phase5/checkpoint_5_*.csv
Running history of decisions and earlier results	docs/PROGRESS_LOG.md
New scripts	scripts/30_clean_v2.py, 31_features.py, 34_claim_tests.py, 34b_survivorship_probe.py, 35_baseline_replication.py, 35b_paired_test.py, 36_pit_universe.py, xs_lib.py
Runs still to do	run_35_baseline_with_lstm.bat, run_33_download_and_model.bat

Sources

- Iskakov et al. (2026), Machine Learning-Based Portfolio Optimization Using Cross-Sectional Signals: Evidence from the NSE Market, ETASR 16(4). etasr.com
- Kotekar, Mohan and Kolukuluri, Can News Sentiment Improve Deep Learning Models for Nifty 50 Index Forecasting?, IEEE Access. idr.nitk.ac.in
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